

MODULE 32 · DOMAIN 3 - CONTINUOUS IMPROVEMENT FOR EXISTING SOLUTIONS

Cost Optimisation of Running Systems

Closing Domain 3. Three kinds of waste, and they need different fixes.

DOMAIN 3 of 4 EXAM WEIGHT 25% MODULE 32 of 44

Three different problems wear the same name. **Unused, oversized, and undiscounted** need different fixes.

The three kinds of waste

KIND	LOOKS LIKE	THE FIX
Unused	Idle instances, unattached volumes, old snapshots, idle load balancers	Delete it
Oversized	Instances at 8% CPU, over-provisioned Lambda memory	Rightsize it
Undiscounted	Steady On-Demand usage with no commitment	Commit to it
Architectural	NAT processing, cross-AZ chatter, egress	Redesign it, as in module 22

ON THE EXAM

Do them in that order. Committing to usage you are about to delete is the most expensive mistake on this list.

Finding the unused

- ◆ **Trusted Advisor** flags idle instances, low-utilisation resources, idle load balancers, unassociated Elastic IPs and over-provisioned Lambda memory.
- ◆ **Unattached EBS volumes and orphaned snapshots** accumulate invisibly, because nothing fails when they are left behind.
- ◆ **S3 Storage Lens** shows where objects and money have accumulated across every account.
- ◆ Check before deleting. A volume detached for a reason, or a snapshot referenced by a launch template, still matters.

ON THE EXAM

Nothing breaks when waste is left behind, which is exactly why it accumulates. Deletion needs a process, not a spring clean.

Utilisation or coverage?

UTILISATION

Am I using what I bought?

- Percentage of commitment consumed
- Low means you bought too much
- Target above roughly 90%
- A multi-day drop is an early warning

COVERAGE

Am I discounting what I run?

- Percentage of eligible usage covered
- Low means paying On-Demand needlessly
- There is no single right number
- Raise it only for stable baseline usage

THE TIEBREAKER

High utilisation with low coverage means buy more. Low utilisation means you already bought too much. They point opposite ways.

Rightsize or commit first?

- ◆ **Committing first** applies a discount immediately across everything eligible, with no engineering work at all.
- ◆ **But rightsizing afterwards shrinks the usage**, and the commitment stays where it was. Utilisation falls and you owe the difference.
- ◆ **Commit to the baseline that will still exist after the planned rightsizing**, rather than to today's inflated usage.
- ◆ Cost Explorer recommendations use 30 to 60 days of history and refresh roughly every 72 hours, so recent changes are not yet reflected.

ON THE EXAM

This is a genuine trade-off, not a rule. What decides it is whether a rightsizing programme is actually going to happen.

Noticing sooner

- ◆ **Cost Anomaly Detection** learns your spending pattern and alerts on deviation, at service, account, tag or cost category level.
- ◆ It is detective. It tells you after the spend has happened, so pair it with budgets and budget actions.
- ◆ **Budget actions** from module eleven are the preventive half: apply a policy or stop resources at a threshold.
- ◆ Monitor at the right granularity. An anomaly in one account is invisible in an organisation-wide total.

ON THE EXAM

A monthly bill review means the average surprise is discovered two weeks after it started. Anomaly detection closes that gap.

Attributing it

- ◆ The Cost and Usage Report is the only source with per-resource, hourly detail. Cost Explorer summarises.
- ◆ Query it with Athena to answer which team, which resource, and which hour, from module eleven.
- ◆ Attribution depends entirely on tags, and cost allocation tags are **not retroactive**. Module eleven again.
- ◆ Showback changes behaviour on its own. Teams that can see their own spend reduce it without being asked.

ON THE EXAM

Optimising centrally has a ceiling. Showing each team its own number moves more spend than any central programme.

Reading the scenario

THE SCENARIO SAYS

Savings Plan utilisation is 70%

Utilisation 99%, coverage 45%

Rightsizing dropped utilisation

A spike was found on the monthly bill

Unattached volumes and old snapshots

Which team caused this increase?

IT IS POINTING AT

Over-committed. You bought too much

Under-committed. Buy more for the baseline

Committed before shrinking the usage

Cost Anomaly Detection

Trusted Advisor, plus a deletion process

CUR with Athena, and tags

ON THE EXAM

Decide first which of the three wastes you are looking at. Unused, oversized and undiscounted have nothing in common but the symptom.

Meridian Freight reports Savings Plans utilisation of 68% and coverage of 82%. The commitment was purchased 14 months ago for a three-year term. Finance asks why the savings are lower than projected.

Compute usage has fallen since a containerisation project moved several workloads to Fargate.

What is the situation, and what should be done?

- ☐ A Over-committed: 32% of the commitment is unused. Grow eligible usage onto the plan, and size the next purchase to the reduced baseline
- ☐ B Under-committed: buy an additional Savings Plan to raise coverage
- ☐ C Cancel the Savings Plan and return to On-Demand
- ☐ D Nothing; 68% utilisation is within normal range

◆ PAUSE AND DECIDE

What is the situation, and what should be done?

- A

Over-committed

Utilisation of 68% means nearly a third of the committed spend buys nothing. The containerisation project shrank the usage under a fixed commitment.
- B

Buy more

Reads coverage and ignores utilisation. Buying more commitment when a third of the existing one is unused makes it worse.
- C

Cancel it

Savings Plans cannot be cancelled. The seven-day return window applies only to a recent purchase, not one 14 months old.
- D

68% is fine

Practice targets above roughly 90%. At 68%, almost a third of every committed dollar is producing no discount at all.

THE SIGNAL

Fargate is eligible for a Compute Savings Plan, so the containerisation should not have reduced utilisation. Unless the plan was EC2 Instance type.

A company reports Savings Plans utilisation of 99% and coverage of 41%. Compute usage has been stable for two years and is forecast to remain so. The finance team believes the Savings Plan strategy is working well and no action is needed.

The uncovered usage is steady, round-the-clock production workload.

What should the solutions architect advise?

- ☐ A Agree; 99% utilisation shows the commitment is correctly sized
- ☐ B Coverage of 41% means most eligible usage is paying On-Demand rates; increase the commitment for the stable baseline
- ☐ C Reduce the commitment, since utilisation cannot exceed 100%
- ☐ D Switch the remaining usage to Spot Instances

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A

 Agree, 99% is good
Utilisation of 99% is excellent and only says the small commitment is fully used. It says nothing about the 59% paying full price.
- B

 Coverage is low; commit more
Stable, round-the-clock, forecast to continue. That is exactly the profile a commitment exists for, and 59% of it is undiscounted.
- C

 Reduce the commitment
Backwards. High utilisation with low coverage means the commitment is too small, not too large.
- D

 Move to Spot
Steady round-the-clock production is the workload least suited to interruption. Spot suits fault-tolerant, flexible work.

THE SIGNAL

Pair this with scenario one. Same two metrics, opposite conclusions, because they measure different things.

A company bought a large three-year Compute Savings Plan in January, sized to its On-Demand spend at the time. In March it completed a rightsizing programme that reduced compute consumption by 30%. Savings Plans utilisation has since sat at 71%.

Both projects were run by different teams.

What went wrong, and what should be done differently next time?

- ☐ A The commitment was sized to pre-rightsizing usage; size commitments to the baseline that will remain after planned changes
- ☐ B The rightsizing programme was a mistake and should be reversed
- ☐ C Convert the Compute Savings Plan to an EC2 Instance Savings Plan
- ☐ D Nothing went wrong; utilisation recovers naturally over a three-year term

◆ PAUSE AND DECIDE

What went wrong, and what should be done differently next time?

- A

Sized to pre-rightsizing usage

Two teams optimised independently and the second undercut the first. Size to the baseline that survives the planned change.
- B

Reverse the rightsizing

Runs oversized instances for three years to justify a commitment. It optimises the metric rather than the bill.
- C

Convert the plan type

Savings Plans cannot be converted between types. Convertible Reserved Instances can be exchanged; Savings Plans cannot.
- D

It recovers naturally

Only if usage grows back to the pre-rightsizing level, which is the opposite of the intent. Nothing recovers on its own.

THE SIGNAL

Two cost programmes run independently can subtract from each other. That coordination failure is the real lesson here.

A misconfigured job began writing 40 TB a day to S3 in a development account on the 3rd of the month. It was discovered on the 14th during a routine bill review, by which time it had cost roughly \$28,000. Budgets are set at the organisation level only.

The development account normally spends about \$900 a month.

What should the solutions architect implement?

- ☐ A Cost Anomaly Detection monitors at account level, with alerts, plus per-account budgets and budget actions
- ☐ B Review the organisation bill weekly instead of monthly
- ☐ C Lower the organisation-level budget threshold
- ☐ D Apply an SCP preventing development accounts from writing to S3

◆ PAUSE AND DECIDE

What should the solutions architect implement?

- A

 Account-level anomaly monitors and budgets
A jump from \$900 to thousands is glaring against that account's own baseline, and invisible in the organisation total. Budget actions can stop it.
- B

 Weekly bill review
Reduces discovery from 11 days to about 7. Still a week of spend, and it still depends on somebody reading a report.
- C

 Lower the org threshold
\$28,000 may be noise in an organisation-wide budget. The granularity is the problem, not the threshold.
- D

 Block S3 in dev
Prevents legitimate development work to stop a misconfiguration. It solves the symptom by removing the capability.

THE SIGNAL

The \$900 baseline is the planted detail. It tells you the anomaly is enormous locally and invisible globally.

An audit across 80 accounts finds 3,400 unattached EBS volumes, 19,000 snapshots older than two years, 240 unassociated Elastic IPs and 60 load balancers with no healthy targets. Estimated waste is \$47,000 a month.

Two previous manual clean-up attempts stalled because nobody could confirm what was safe to delete.

What should the solutions architect recommend?

- ☐ A Use Trusted Advisor and Storage Lens to identify candidates, then a tag-based process: tag, notify owners, wait, then delete automatically
- ☐ B Delete everything older than two years immediately
- ☐ C Ask each account owner to review and clean up their own account
- ☐ D Accept the waste, since \$47,000 is small relative to total spend

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 Identify, tag, notify, wait, delete
Addresses why the previous attempts stalled. The waiting period converts silence into consent and makes deletion automatic rather than a decision.
- B

 Delete immediately
A snapshot referenced by a launch template, or a volume detached for a forensic hold, breaks something. This is why attempts one and two stalled.
- C

 Ask the owners
Exactly what failed twice. Owners cannot confirm what is safe either, and the request competes with their own work.
- D

 Accept it
Over half a million pounds a year, growing, for resources nobody uses. It also normalises accumulation.

THE SIGNAL

Two failed attempts is the planted detail. It tells you the obstacle is decision-making, not identification.

The monthly bill rose by \$60,000. Finance asks which team caused it. Cost Explorer shows the increase is in EC2 and data transfer across three accounts, but cannot attribute it to a team or workload.

Resources carry a Team tag, activated as a cost allocation tag 18 months ago.

What should the solutions architect do?

- ☐ A Query the Cost and Usage Report in Athena, grouped by the Team tag and by resource, at hourly granularity
- ☐ B Filter Cost Explorer by the Team tag
- ☐ C Enable Cost Anomaly Detection and wait for the next occurrence
- ☐ D Ask each of the three account owners to investigate

◆ PAUSE AND DECIDE

What should the solutions architect do?

- A** CUR in Athena, by tag and resource
The only source with per-resource hourly detail. The tag was activated 18 months ago, so the data exists for this period.
- B** Filter Cost Explorer
The near miss. Grouping by tag works, and Cost Explorer will not reach the individual resource and hour that identify the change.
- C** Anomaly Detection and wait
Worth enabling for next time, and it answers nothing about the \$60,000 already spent. Finance asked about this month.
- D** Ask the owners
Delegates an investigation the data can answer directly, to people who will look at the same Cost Explorer view.

THE SIGNAL

The tag was activated 18 months ago, which is the detail that makes this answerable. Activation is not retroactive.

What to carry out of this module

- ◆ **Unused, oversized, undiscounted.** Delete, resize, commit. Doing them out of order commits you to waste.
- ◆ **Low utilisation means you bought too much. Low coverage means you bought too little.** They point opposite ways.
- ◆ **Size a commitment to the baseline that survives your planned rightsizing,** not to today's usage.
- ◆ Monitor anomalies at a granularity where they are visible. A big spike in a small account vanishes in an org total.

ON THE EXAM

When a clean-up has stalled twice, the obstacle is decision-making. Tag, notify, wait, then delete automatically.